

HONORS PHYSICS: TACTICAL HIGHWAY
INTERCEPT

Names: _____

Period: 0

Date: _____

UNIT 2: PIECEWISE 1D MOTION, REST INTERVALS & PURSUIT
VELOCITY

SURVEILLANCE DOSSIER: MULTI-STAGE RECONNAISSANCE TRACKING

Highway traffic sensors and an overhead observation satellite track a suspect vehicle heading eastbound on a straight desert highway starting from coordinate $x = 0\text{ m}$ at $t = 0\text{ s}$. The target's journey consists of three distinct kinematic legs:

- **Leg 1 ($t = 0$ to 20 s):** Drives at a steady speed of **15.0 m/s**.
- **Leg 2 ($t = 20$ to 50 s):** Pulls into an emergency turnout / rest stop and remains **completely stationary ($v = 0\text{ m/s}$)** for 30 seconds.
- **Leg 3 ($t = 50$ to 90 s):** Re-enters the highway and accelerates to a new uniform cruise speed of **25.0 m/s** for 40 seconds.

A Highway Patrol Interceptor is stationed at coordinate $x = 0\text{ m}$. Due to dispatch verification, the cruiser **cannot roll until $t = 30$ seconds**. Once dispatched, it travels at a single uniform speed.

PART 1: TARGET VEHICLE SEGMENT TELEMETRY BREAKDOWN

DISPLACEMENT CALCULATIONS

Leg 1: Initial Highway Cruise (0 to 20 s)

- Velocity: $v_1 = +15.0\text{ m/s}$ • Duration: $\Delta t_1 = 20\text{ s}$
- Displacement: $\Delta x_1 = v_1 \cdot \Delta t_1 = (15.0)(20) = +300\text{ m}$
- Coordinate at $t = 20\text{ s}$: (20 s, 300 m)

Leg 2: Rest Stop Turnout (20 to 50 s)

- Velocity: $v_2 = 0.0\text{ m/s}$ (At Rest) • Duration: $\Delta t_2 = 30\text{ s}$
- Displacement: $\Delta x_2 = 0\text{ m}$
- Coordinate at $t = 50\text{ s}$: (50 s, 300 m)

Calculate Target Leg 3 Displacement & Final Position:

Duration: $\Delta t_3 = 90\text{ s} - 50\text{ s} = 40\text{ s}$ | Velocity: $v_3 = +25.0\text{ m/s}$
Displacement: $\Delta x_3 = (v_3) \cdot (\Delta t_3) = \rule{1.5cm}{0.4pt} = \rule{1.5cm}{0.4pt}\text{ meters}$
Final Coordinate at $t = 90\text{ s}$: (90 s, $\rule{1.5cm}{0.4pt}\text{ m}$)

PART 2: HIGHWAY PATROL PURSUIT STRATEGIES

CRUISER DISPATCH: $T = 30\text{ S}$

Strategy Alpha: Intercept the target while it is still stationary at the rest stop (at $t = 50\text{ s}$):

Available Time: $\Delta t = 50\text{ s} - 30\text{ s} = 20.0\text{ seconds}$ | Required Position: $x = 300\text{ meters}$
Required Cruiser Constant Speed: $v_{\text{alpha}} = \Delta x / \Delta t = \rule{1.5cm}{0.4pt} = \rule{1.5cm}{0.4pt}\text{ m/s}$

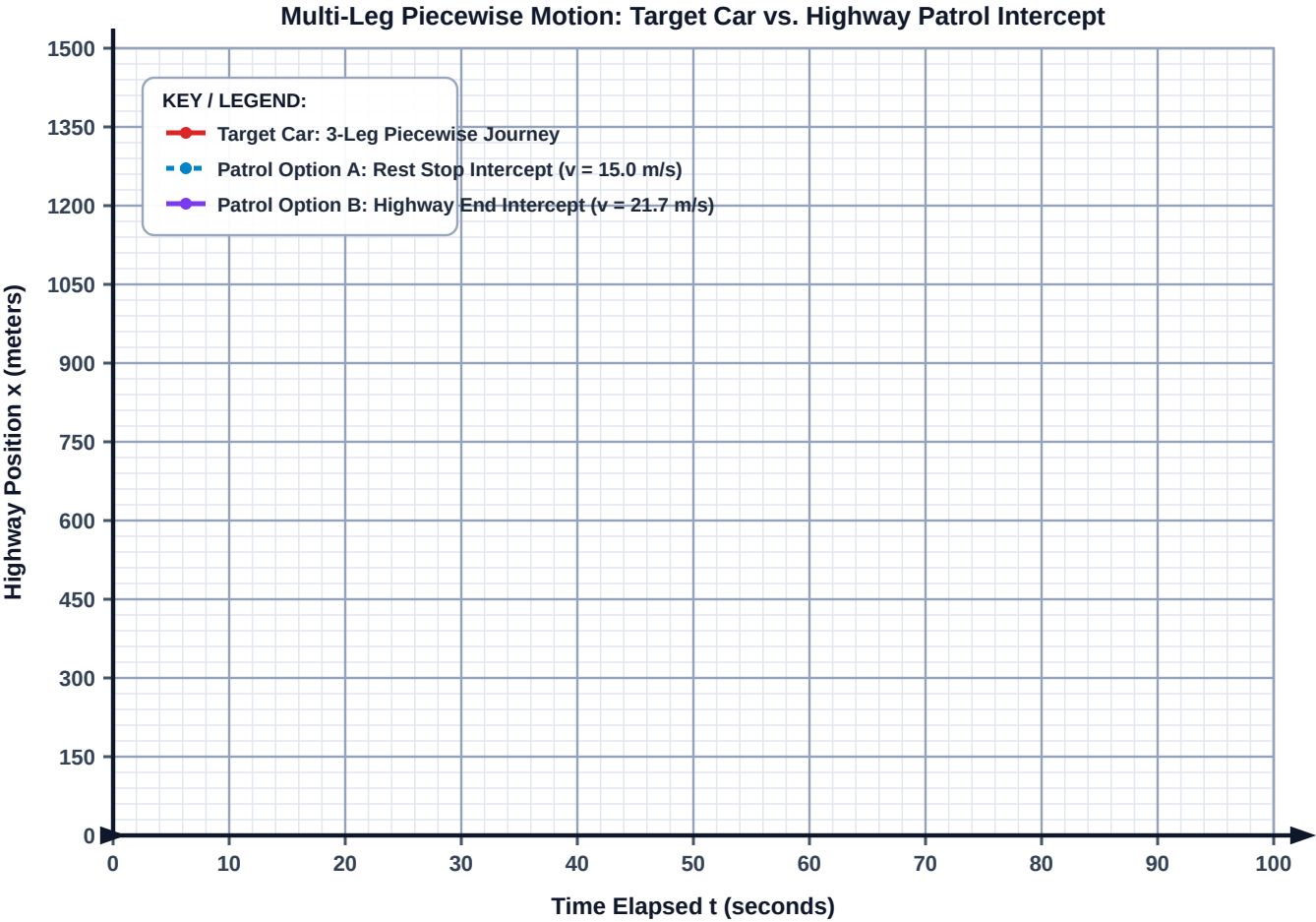
Strategy Beta: Intercept the target at the exact conclusion of Leg 3 (at $t = 90\text{ s}$):

Available Time: $\Delta t = 90\text{ s} - 30\text{ s} = 60.0\text{ seconds}$ | Required Position: $x = 1,300\text{ meters}$
Required Cruiser Constant Speed: $v_{\text{beta}} = \Delta x / \Delta t = \rule{1.5cm}{0.4pt} = \rule{1.5cm}{0.4pt}\text{ m/s}$

PART 3: MASTER HIGHWAY TRAJECTORY GRAPH & ANALYSIS

PLOTTING MULTI-SEGMENT X-T PROFILES • INTERSECTION POINTS • AVERAGE VS. INSTANTANEOUS SPEED

HIGH-PRECISION
GRAPHING



PART 4: GRAPHICAL ANALYSIS & SPEED COMPARISON

HONORS DEBRIEF

1. Slope Meaning on Position-Time Graph:

Explain what the visual slope of the line represents during Leg 2 ($t = 20 \text{ s}$ to 50 s). How can a patrol officer tell instantly from an x - t graph that the target is stationary without looking at numbers?

2. Target Vehicle Overall Average Speed vs. Instantaneous Speeds:

Calculate the target's **Overall Average Speed** across the entire 90-second mission (Total Distance / Total Time). Explain why this value is lower than both Leg 1 (15 m/s) and Leg 3 (25 m/s):

3. Speed Limit Constraint Analysis:

The highway patrol interceptor has a maximum vehicle speed cap of **20.0 m/s (45 mph)**. Which of the two intercept strategies (Alpha or Beta) is physically possible? Defend with calculations: